GREENOPS DIGITAL INFRASTRUCTURE AUDIT PACK

Statutory Scope 2 Compute Disclosures & Cryptographic Provenance Ledger

Tenant: enterprise-logistics-uk | **Generated:** 2026-10-22 21:09:58 UTC | **Assurance Framework:** ISAE 3410 / ISO 14064-2

1. Executive Energy & Emissions Summary

Metric	Value	Accounting Standard / Method
Total Workloads Processed	5	Batch Compute Executions
Total Energy Consumed	124.50 kWh	Direct Hardware & Cluster Metering
Statutory Scope 2 Emissions	19.05 kg CO2e	UK SECR / GHG Protocol (NESO AEF Location-based)
Consequential Carbon Displaced	6.66 kg CO2e	Elexon BMRS Marginal Emissions Factor (MEF) Avoidance
Estimated Spot Cost Savings	£4.61	Wholesale Grid / Off-Peak Spot Shift Arbitrage
SLA Continuity Overrides	0 (0.0%)	Zero-Downtime Pipeline Safety Invariant Active

2. Non-Repudiable Settlement Ledger Blocks (Merkle Provenance)

Settlement Window (UTC)	Jobs	kWh	Merkle Root Hash (SHA-256)
2026-10-22 22:09:58	5	100.00	2088ac0c5b422b2ebe0c7ab6cf3180b3...

3. Statutory Methodology & Auditor Notes

Statutory SECR Scope 2 Compliance: Calculated using location-based Average Emissions Factors (AEF) published by National Energy System Operator (NESO) at 30-minute settlement granularity. Satisfies UK Companies Act 2006 (Strategic Report and Directors' Report) Regulations 2018.
Consequential Avoidance Accounting: Avoided carbon is calculated against an explicit ISO 14064-2 project baseline counterfactual, utilizing short-run Marginal Emissions Factors (MEF) derived from active Elexon BMRS gas peaking generation. Not to be conflated with statutory location disclosures.
Cryptographic Tamper-Evidence: Every settlement window commits an immutable Merkle tree root hash. Individual job inclusion can be independently verified via inclusion proof paths without exposing underlying proprietary pipeline code.